OBJECTIVE

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FOR FINAL TERM (FALL 2021,22)

Course Code: MTH601

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MSC MATHEMATICS



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MTH601 Final Term Papers By Waqar (File 1)

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15/March/2022

MTH601 Operations Research BC08040074

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

The cost matrix in assignment problem is always diagonal matrix

VuAnswers.com

Answer (Please select your correct option)

- square matrix — correct
- identity matrix
- zero matrix

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

VuAnswers.com

Answer (Please select your correct option)

- Subtraction — correct
- Division
- Multiplication
- Addition

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

During a replacement if the value of money decreases at the rate of 3% then the present worth factor of unit amount to be spent after one year is given by

$v = \frac{1}{1+i} = \frac{1}{1+3\%} = \frac{1}{1.03} = 0.9708$

VuAnswers.com

Answer (Please select your correct option)

- 0.25
- 0.333
- 0.9708 — correct
- 4

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

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Answer (Please select your correct option)

- $3n$ — correct
- $3n!$
- $2\times3=6$
- 2

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

If the money carries a rate of interest of 12% per year, the present worth factor of one rupee due in one year is

$\frac{1}{1+i} = \frac{1}{1.12} = 0.89285$

VuAnswers.com

Answer (Please select your correct option)

- 0.08333
- 0.89285 — correct
- 0.0769
- 13

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Degeneracy in a 5×6 transportation problem occurs when the number of occupied cell is less than

$m+n-1$
$6+5-1=10$

VuAnswers.com

Answer (Please select your correct option)

- 20 but greater than 10
- 10 — correct
- Zero
- Infinity

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

We go in probabilistic replacement model when period between installation and failure is

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Answer (Please select your correct option)

- varying exponentially
- varying linearly — correct
- Constant
- is not constant (varying arbitrarily)

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Any set of non negative allocations ($X_{ij} > 0$) which satisfies the row and column sum is called a _____ solution.

VuAnswers.com

Answer (Please select your correct option)

- feasible
- non basic feasible
- basic infeasible
- optimal — correct

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

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Answer (Please select your correct option)

- Degenerate — correct
- Multiple Solutions
- Non degenerate
- Alternative Optima

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following cost table:

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

VuAnswers.com

Answer (Please select your correct option)

- 10
- 15
- 25
- 5

correct

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Source	S1	5	1	2	25
	S2	7	2	4	10
	S3	3	3	5	15
	Demand	10	20	20	

VuAnswers.com

Answer (Please select your correct option)

- 10
- 15
- 25
- 5

correct

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

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Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

correct

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

	S3	3 10	3	5	15
	Demand	10	20	20	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

VuAnswers.com

Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

VuAnswers.com

Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

We go in probabilistic replacement model when period between installation and failure is

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Answer (Please select your correct option)

- varying exponentially
- varying linearly

correct

- Constant
- is not constant (varying arbitrarily)

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If the mean arrival and mean service rates are **4** and 7 respectively in a queue then expected waiting time in the system is

$\frac{1}{\mu-\lambda} = \frac{1}{7-4} = \frac{1}{3}$

VuAnswers.com

Answer (Please select your correct option)

- $\frac{1}{3}$ correct
- 3
- 28
- $\frac{7}{4}$

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

In a bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes only 10 minutes for serving a customer on an average, then the service rate " μ" = _____.

VuAnswers.com

Answer (Please select your correct option)

- 6 per hour correct
- 4 per hour
- 10 per hour
- 1/6 per hour

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A duplicating machine maintained for office use is used and operated by people in the office who need to make copies. Since the work to be copied varies in length (number of pages of the original) and copies required, the service rate is randomly distributed, the arrival rate is 5 per hour and the service rate is 10 per hour then the equipment utilization " ρ" is equal to

VuAnswers.com

Answer (Please select your correct option)

- 0.50 correct
- 0.20
- 5
- 2

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the average rate " λ " is

VuAnswers.com

Answer (Please select your correct option)

- 1/8 = 0.125 — correct
- 8
- 4
- ¼ = 0.25

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the mean service time " μ " is

VuAnswers.com

Answer (Please select your correct option)

- ½ = 0.5 — correct
- 4
- ¼ = 0.25
- 2

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If "N_i" be the Number of replacement made at the end of the ith week and "P_j" be the probability of failure during the ith week, then N_i = ----------.

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Answer (Please select your correct option)

- N_0P_1 — correct
- N_1P_1
- N_0P_0
- N_1P_2

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

The present worth of a rupee to be spent after a year is denoted by v and given by

VuAnswers.com

Answer (Please select your correct option)

- $v = 1/(1+r)$ correct
- $v = (1+r)$
- $v = (1+r)/10$
- $v = (1/r)$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Formula for a geometric series "$x + vx + v^2x + \ldots + v^{n-1}x$" is

VuAnswers.com

Answer (Please select your correct option)

- $P(n) = x\,\frac{1-v^n}{1-v}$ correct
- $P(n) = \frac{1-v^n}{1-v}$
- $P(n) = x\frac{1-v}{1-v^n}$
- $P(n) = \frac{1-v}{1-v^n}$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Product of 'item cost' and 'ordered item' is

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Answer (Please select your correct option)

- Crash cost
- Cost period correct
- Set up cost
- Shortage cost

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Standard Deviation (S.D.) = _____ MAD

VuAnswers.com

Answer (Please select your correct option)

- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{2}{\pi}}$
- $\sqrt{\frac{\pi}{2}}$ correct
- $\sqrt{\frac{3\pi}{2}}$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

MAD = --------- S.D.

VuAnswers.com

Answer (Please select your correct option)

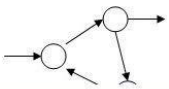
- $\sqrt{\frac{2}{\pi}}$ correct
- $\sqrt{\frac{\pi}{2}}$
- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{3\pi}{2}}$

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The following network is an example of



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Answer (Please select your correct option)

- Redundancy
- Dangling correct
- Cycling
- Dummy

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

For any activity backward pass computations provide its

VuAnswers.com

Answer (Please select your correct option)

- Earliest start times
- Latest start times **correct**
- Moderate start times
- Completion time

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Best possible time estimate that a given activity would take under normal conditions which often exist, is called

VuAnswers.com

Answer (Please select your correct option)

- Most Likely time estimate
- Pessimistic time estimate **correct**
- Smallest time estimate
- Activity time estimate

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

In PERT, activity time estimates are distributed according to

VuAnswers.com

Answer (Please select your correct option)

- Beta Distribution **correct**
- Normal Distribution
- Poisson distribution
- Binomial Distribution

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Solution region for the constraint $y \leq 0$ is the

VuAnswers.com

Answer (Please select your correct option)

- Half plane below y-axis
- Half plane below the line: $y = 0$
- Set of all those points where ordinates are non-positive
- All are equivalent

correct

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

In a linear Programming Problem (LPP), which of the following must be hold?

VuAnswers.com

Answer (Please select your correct option)

- Only objective function is linear
- Both objective function and constraints are linear

correct

- Only constraints needs to be linear
- At least one of objective function or constraint should be linear

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If a company manufacture 'x' units of product 'A' and 'y' units of 'B' with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?

VuAnswers.com

Answer (Please select your correct option)

- $z = 15xy$
- $z = 5x - 3y$
- $z = 3x - 5y$
- $z = 5x + 3y$

correct

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ variable is chosen by examining the cost coefficents in the objective function.

VuAnswers.com

Answer (Please select your correct option)

- entering — correct
- leaving
- positive slack
- negative slack

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

While solving a linear programming problem by big M – Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

VuAnswers.com

Answer (Please select your correct option)

- neagtive slack
- positive slack
- entering
- artificial — correct

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

In the big-M method, if the introduced _____ variables do not leave the basis in the final iteration, then this indicates that the give linear programming problem cant be optimized.

VuAnswers.com

Answer (Please select your correct option)

- entering
- positive slack
- negative slack
- artificial — correct

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Zero valued artificial variables may appear as _____ variables in the final solution, when one or more of the original constraints equations is redundant.

VuAnswers.com

Answer (Please select your correct option)

- non basic
- basic

correct

- slacks
- surplus
- artificial

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

By using Two phase method to solve a linear programming problem, in phase I, a new objective function is formed by assigning on left hand side, zero to every original variable (including slack and surplus variables) and -------- to each of the artificial variables.

VuAnswers.com

Answer (Please select your correct option)

- M
- –M

correct

- +1
- –1

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

For the linear programming problem;

$\text{Max} Z = 2x + 3y$

Subject to

$$\left.\begin{array}{l} x \geq 2 \\ y \leq 3 \\ x, y \geq 0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} x - s_1 + A = 2 \\ y + s_2 = 3 \\ x, y, s1, s2, A \geq 0 \end{array}\right\}$$

VuAnswers.com

Answer (Please select your correct option)

- $\text{Max} Z = 2x + 3y + A$
- $\text{Min} Z = 2x + 3y + A$
- $\text{Max} Z = A$
- $\text{Min} Z = A$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

For the linear programming problem;
Max$Z = 2x + 3y$
Subject to

$$\left.\begin{array}{l} x \geq 2 \\ y \leq 3 \\ x, y \geq 0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} x - s_1 + A = 2 \\ y + s_2 = 3 \\ x, y, s1, s2, A \geq 0 \end{array}\right\}$$

VuAnswers.com

Answer (Please select your correct option)

- Max$Z = 2x + 3y + A$
- Min$Z = 2x + 3y + A$
- Max$Z = A$
- Min$Z = A$

correct

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

$$\left.\begin{array}{l} y \leq 3 \\ x, y \geq 0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} y + s_2 = 3 \\ x, y, s1, s2, A \geq 0 \end{array}\right\}$$

Which of the following is associated objective function of the 1st phase?

VuAnswers.com

Answer (Please select your correct option)

- Max$Z = 2x + 3y + A$
- Min$Z = 2x + 3y + A$
- Max$Z = A$
- Min$Z = A$

correct

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The inequality $2x + 3y \geq 18$ is equivalent to

VuAnswers.com

Answer (Please select your correct option)

- $2x + 3y \leq -18$
- $-2x - 3y \geq 18$
- $-2x - 3y \geq -18$
- $-2x - 3y \leq -18$

correct

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?

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Answer (Please select your correct option)

- Transportations models
- Degenerate Linear models

correct

- Non-degenerate Linear models
- Dual or unbounded linear models

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Transportations models consist of ------- like the production centers and --------- which may be the sales centers.

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Answer (Please select your correct option)

- (sinks, sources)
- (sources, sinks)

correct

- (origins, sources)
- (sinks, destinations)

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

To convert the transportation problem into a maximization model we have to----------.

VuAnswers.com

Answer (Please select your correct option)

- write the inverse of the coefficient matrix
- multiply the feasibility condition by −1
- multiply the coefficient matrix by −1
- We can't convert the transportation problem into a maximization problem, as it is basically a minimization prob

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Operations Research (MTH601)

Question: 1 (Marks: 1) Attempted Questions: 0 Total Questions: 52

Under the individual replacement policy an item is replaced on its

.....

Choices:

Success

.....

Failure

.....

Maximum utilization

.....

Minimum utilization

.....

TIME LEFT

112

Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 2 (Marks: 1) Attempted Questions: 1 Total Questions: 52

During a replacement if the value of money decreases at the rate of 3% then the present worth factor of unit amount to be spent after one year is given by

Choices:

0.25

0.333

0.9708

4

TIME LEFT
112

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

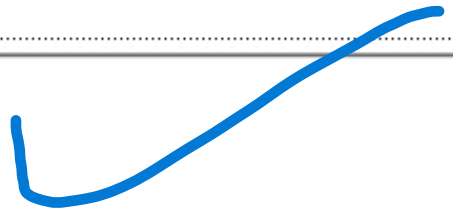
Choices:

- $3n$
- $3n!$
- $2 \times 3 = 6$
- 2

The name of the method in getting the optimum assignment is

Choices:

- Simplex method
- North west corner method
- Vogal approximation method
- Hungarian method



Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 5 (Marks: 1) Attempted Questions: 4 Total Questions: 52

If the mean arrival and mean service rates are **4** and 7 respectively in a queue then expected waiting time in the system is

Choices:

1/3



28

3

7/4

TIME LEFT

109

Transportation technique or the simplex method cannot be used to solve the assignment problem because of

Choices:

- Degeneracy
- Involvement of dummy activities
- Looping among the entries of assignment matrix
- Inequilibrium between demand and supply

Question: 6 (Marks: 1) Attempted Questions: 5 Total Questions: 52

The representation of reality in some physical form or in some form of mathematical equations may be discussed under the topic entitled as

Choices:

- Transportation problems
- Game theory
- Simulation
- Replacement problems

Question: 8 (Marks: 1) Attempted Questions: 6 Total Questions: 52

Search and cutting methods are used to solve following types of problems

Choices:

- Transportation
- Assignment
- Integer Programming
- Queuing problems

CPM can help you figure out EXCEPT:

Choices:

how long your complex project will take to complete

which activities are "critical," meaning that they have to be done on time or else the whole project will take longer

what is the most costly way to speed up the project

you can try to speed up the project



Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 10 (Marks: 1) Attempted Questions: 8 Total Questions: 52

The Vogel Approximation method is an iterative procedure for computing a _____ solution of the transportation problem.

Choices:

basic feasible

non basic feasible

basic infeasible

optimal

TIME LEFT 107

Question: 11 (Marks: 1) Attempted Questions: 9 Total Questions: 52

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

Choices:

- Degenerate
- Multiple Solutions
- Non degenerate
- Alternative Optima

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10
	S3	3	3	5	15
	Demand	10	20	20	

Choices:

10

15

25

5

Question: 14 (Marks: 1) Attempted Questions: 11 Total Questions: 52

The name of the method in getting the optimum assignment is

Choices:

- Simplex method
- North - West corner method
- Vogel's approximation method
- Hungarian method

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

Choices:

- Subtraction
- Division
- Multiplication
- Addition

In assignment problem, each job requires exactly one resource.

Choices:

True

False

May not be true

If "N_i" be the Number of replacement made at the end of the ith week and "P_j" be the probability of failure during the ith week, then N_2 = ----------.

Choices:

$N_0P_2 + N_1P_1$

$N_0P_1 + N_1P_2$

$N_0P_2 + N_1P_2$

$N_1P_2 + N_1P_1$

In replacement Models, Capital Cost = (Item / Machine Cots) – (----------).

Choices:

- Resale Cost
- Running Cost
- Maintenance Cost
- Operating Cost

Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, we have the relations

I. TS ≤ FS
II. INDS ≤ FS
III. FS ≤ TS

Then which relation is true

Choices:

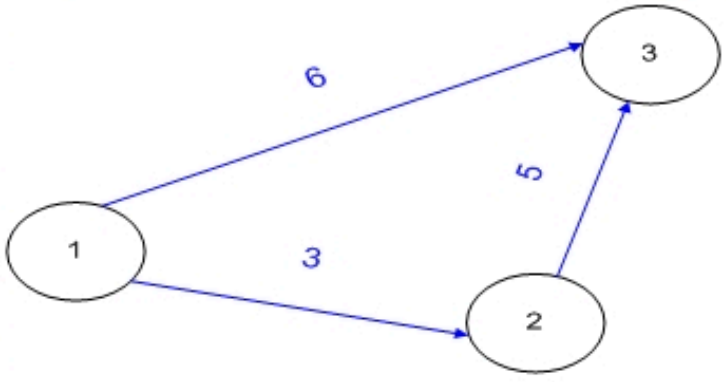
- Only I
- Only II
- Only III
- Only I & II
- Only II & III

Which of the following give the excess of available time over the activity time when all jobs start as early as possible?

Choices:

- Total float
- Free float
- Independent float
- Early finish times

Early finish time of the event '3' in the following project is -------.



Choices:

14 days

2 days

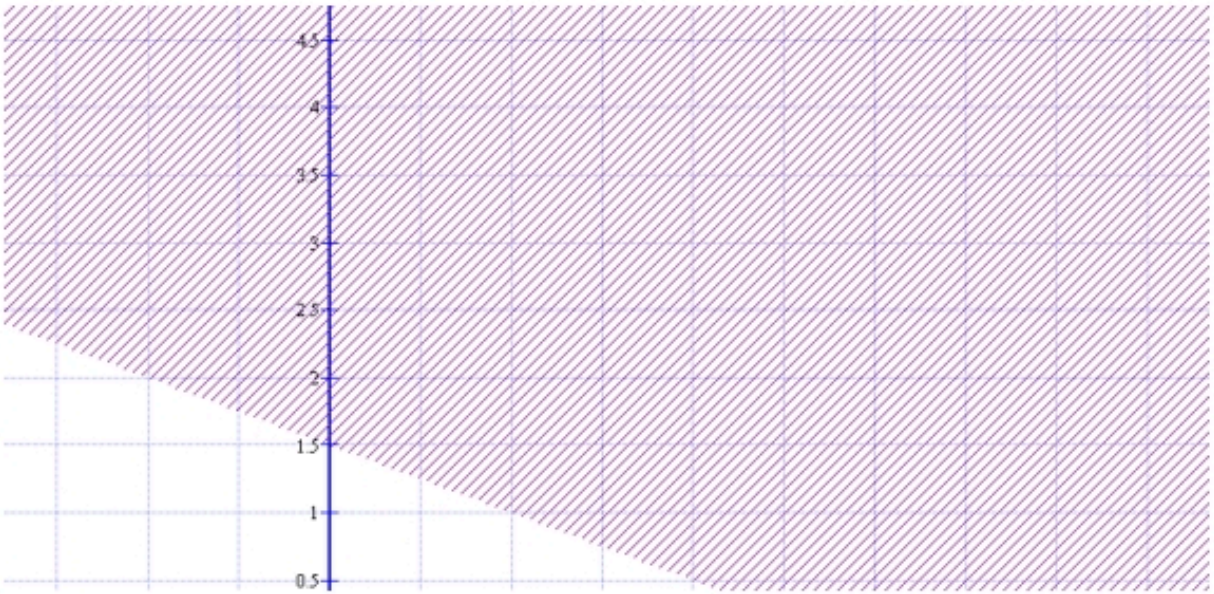
8 days

6 days

Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 25 (Marks: 1) Attempted Questions: 16 Total Questions: 52



Choices:

$x+2y \leq 3$

$x+2y \geq 3$

$x-2y \leq 3$

$x-2y \geq 3$

N.C

TIME LEFT

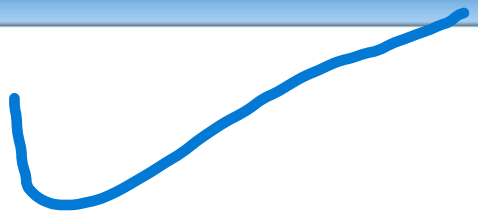
102

Question: 26 (Marks: 1) Attempted Questions: 16 Total Questions: 52

It is a property of simplex method that there always exist infinite numbers of basic feasible solutions.

Choices:

True



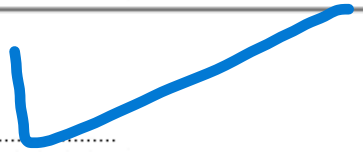
False

Question: 27 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If the objective function of a linear programming problem needs further improvement then which of the following will have to proceed?

Choices:

- a new decision variable to be entered and other decision variable to leave the basis
- a new slack variable to be entered and other slack to leave the basis
- a new non-degenerate variable to be entered and other non-generate to leave the basis
- a new basic variable to be included and other basic variable to leave the basis



The constrained $3x_1 + 2x_2 \leq 18$ in standard form can be written as:
(Where R is artificial variable and S_1 is Slack variable)

Choices:

- $3x_1 + 2x_2 + S_1 + R = 18$
- $3x_1 + 2x_2 + S_1 - R = 18$
- $3x_1 + 2x_2 - S_1 + R = 18$
- $3x_1 + 2x_2 + S_1 = 18$

If no _____ variable appears in the basis and the optimality conditions are satisfied, then the current solution will be optimal one.

Choices:

- artificial
- non basic
- slack
- surplus

Min. Z = 0 and no artificial variable appears in the basic variables. _____ solution to original problem has been found and we proceed to phase II method.

Choices:

A basic feasible

An optimal

An infeasible

If the problem has a _____ solution such that the minimum value of the objective function is zero along with zero values of artificial variables. Then proceed to phase II method.

Choices:

- unbounded
- feasible
- non feasible
- de-generate feasible solution

Question: 32 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If the degeneracy arises in the initial stage, then which of the following is true?

Choices:

- One of the basic variables is zero.
- More than one variable is eligible to leave the basis.
- Tie in the ratio of each row,by taking right hand side of each row and dividing by the corresponding element of the key column.
- Iterative Cycling of basic solution exists without reaching the optimality.

Question: 33 (Marks: 1) Attempted Questions: 16 Total Questions: 52

For a linear programming problem, the Unboundedness is related to which of the following?

Choices:

- Solution space only
- Objective value only
- Neither to solution space nor to objective values
- Both solution space and objective value

Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	S_1	S_2	S_3	R.H.S
Z	-4	-3	-2	0	0	0	0
x_1	0	0	1	1	0	0	8
S_2	-2	1	3	0	1	0	9
S_3	0	2	1	0	0	1	3

Then this is the indication that

Choices:

- Problem has Unbounded Solutions
- Problem has Infeasible Solutions
- Problem has temporarily degenerate
- Problem has Multiple Optimal Solutions

Consider the following simplex table corresponding to the maximization problem

Basic	x_1	x_2	x_3	x_4	A_1	A_2	R.H.S
Z	0	0	M + 1/2	M	16M – 11	0	-0.6M – 5.7
x_1	1	0	5	0	-1	0	3
x_2	0	1	-5	0	3	0	9
A_2	0	0	-1	-1	-0.6	1	0.6

This is the indication that problem has----------.

Choices:

- infeasible Solution
- temporarily Degenerate Solution
- unbounded Solution
- degenerate Solution

In the Simplex table, if the coefficient of non-basic variable says x_1 in the z-row is zero, then it indicates that value of z ----------.

Choices:

- increase
- decreases
- does not change
- becomes infinity

A primal's constraint equation produces _____ dual variable.

Choices:

non basic

an unrestricted

artificial

basic

Question: 38 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is -------.

Choices:

- Zero
- Maximum
- Minimum
- Average

Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 39 (Marks: 1) Attempted Questions: 16 Total Questions: 52

Many decision making problems involve a process that takes place in multiple stages in such a way that at each stage, the process is dependent on the strategy chosen. Such types of problems are called

Choices:

- Linear Programming problems
- Integer programming problems
- Dynamic programming problems
- Assignment problems

TIME LEFT 101

Virtual University Overseas Examination System V2.0 (x64)

Operations Research (MTH601)

Question: 40 (Marks: 1) Attempted Questions: 16 Total Questions: 52

If shortage cost is infinity, then .....

Choices:

no shortages are allowed

no inventory carrying cost is allowed

Ordering cost is zero

Purchase cost = Carrying cost

TIME LEFT 101

Mth601 Quiz#2

Lecture (23-28)

Due Date (7-9) feb2022

- Mth601-Quiz-2 riz mughal 6feb2021 (pg32)
- Mth601 Quiz#2(lec23 to 28)(7Feb2022) made by stylish (pg72)
- Mth601 Quiz2 feb2022 (pg70)
- MTH601 SS Quiz 2 video23-28+others feb0822 (pg111)
- 3 groups quiz unlimited

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Question # 1 of 10 (Start time: 10:05:12 PM, 24 July 2021) Total Marks: 1

In Simplex standard table to solve an LP problem of Minimization, we choose the candidate for entering variable in---------.

Minimization k liyee most positive

Maximization k liyee most negative

Select the correct option

- Constraint with most positive coefficient
- Objective function with most positive coefficient ✓

pg 88

- Constraint with most negative coefficient
- Objective function with most negative coefficient

Click to Save Answer & Move to Next Question

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MTH601: Quiz 2

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

While solving a linear programming problem by big M-Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- negative slack
- positive slack
- entering
- artificial

(Lecture 23, Pg.94)

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MTH601: Quiz 2

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

In the big-M method, if the introduced _____ variables do not leave the basis in the final interaction, then this indicates that the give linear programming problem cant be optimized.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- entering
- positive slack
- negative slack
- artificial

(Lecture 23, Pg.94)

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

While solving an LP by two phase method, an objective function of 1st phase is always of ____

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- maximization
- minimization (Lecture 24, Pg.101)
- dependent on original objective function
- None of these

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Question # 3 of 10 (Start time: 03:33:59 PM, 06 February 2021)

If the primal constraint is originally in equation form, the corresponding dual variable is not unrestricted.

Select the correct option

- ○ True
- ◉ False ✓

RIZ MUGHAL

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Question # 6 of 10 (Start time: 03:34:45 PM, 06 February 2021) **Total**

While facing degeneracy in solving an LP problem,then the further iterations always assure the non-negative optimal solution.

Select the correct option

- ○ True
- ◉ False

RIZ MUGHAL

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MTH601:Quiz

Question # 7 of 10 (Start time: 03:34:59 PM, 06 February 2021)

Dual of a Dual of Dual is------.

Select the correct option

- Dual
- Primal
- Double dual
- Single dual

RIZ MUGHAL

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Question # 1 of 10 (Start time: 05:41:04 PM, 06 February 2021) **Total Mark**

If x_1 unrestricted variable and $x_2 = 0$, then standard primal form of objective function of Z. Where Max $Z = 7x_1 + 89x_2$.

Select the correct option

Reload Math Equation

○	Max $Z = 7(x_1^+ + x_1^+) + 89x_2$
○	Max $Z = 7(x_1^+ + x_1^-) + 89x_2$
◉	Max $Z = 7(x_1^+ - x_1^-) + 89x_2$
○	Max $Z = 7(x_1^- + x_1^-) + 89x_2$

RIZ MUGHAL

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Question # 2 of 10 (Start time: 05:41:24 PM, 06 February 2021)

Which of the following will be an example of degenerate basic feasible solution for an LP problem?

Select the correct option

- ○ (2,3,-1)
- ◉ (0,2,1)
- ○ (2,1,3)
- ○ (-1,2,0)

RIZ MUGHAL

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Question # 3 of 10 (Start time: 05:41:40 PM, 06 February 2021) **Total**

While solving a Linear Programming problem by Simplex Method, an inequality constraint is transformed into strict equality by the inclusion of ---------- variable.

Select the correct option

- [x] slack
- [ ] decision
- [ ] artificial
- [ ] basic

RIZ MUGHAL

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Question # 5 of 10 (Start time: 05:42:11 PM, 06 February 2021)

The dual of a dual problem yields the original primal.

Select the correct option

- (●) True
- ( ) False

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Question # 6 of 10 (Start time: 05:42:26 PM, 06 February 2021) **Total Mark**

If a variable in the Primal is unrestricted in sign, then the corresponding constraint in the dual will be of -------- type and vice versa.

Select the correct option

- ○ >=
- ○ <=
- ◉ = ✓
- ○ none of these

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Question # 3 of 10 (Start time: 05:46:27 PM, 06 February 2021)

Degenerate solutions correspond to---------- variables.

RIZ MUGHAL

Select the correct option

- ○ Dual
- ○ Non-basic
- ◉ Basic
- ○ Unbounded

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Question # 10 of 10 (Start time: 05:48:17 PM, 06 February 2021) **Total Marks: 1**

While solving an LP problem, if the candidate for a leaving variable becomes zero, then solution is said to be --------.

Select the correct option

- ○ optimal solution
- ○ infeasible solution
- ○ non-degenerate solution
- ◉ degenerate solution

RIZ MUGHAL

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Question # 4 of 10 (Start time: 09:43:41 PM, 09 February 2022)

In Simplex standard table to solve an LP problem of Maximization, we choose the candidate for entering variable in-------.

Select the correct option

○	Objective function with most negative coefficient
○	Constraint with most positive coefficient
◉	Objective function with most positive coefficient
○	Constraint with most negative coefficient

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MTH601 – Operations Research (Quiz No 2) **Quiz Start Time: 10:28 PM, 09 February 2022**

Question # 9 of 10 (Start time: 10:36:27 PM, 09 February 2022) **Total Marks:** 1

When we have $\geq$ type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

Select the correct option

Reload Math Equations

○	artificial, Two phase
○	artificial, Big M
●	artificial and slack, Big M or Two phase
○	slack, Big M

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Question # 9 of 10 (Start time: 08:59:32 PM, 09 February 2022) **Total Marks:** 1

In Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column entries should be zero except----.

Select the correct option

○	key row(pivot)entry
○	row with maximum ratio on RHS column
○	1st row entry
○	last row entry

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Question # 5 of 10 (Start time: 10:19:31 PM, 09 February 2022) **Total Marks:** 1

In the Simplex method, if all the elements in columns of the ratio by taking the RHS of each row and dividing by the corresponding element of the key column, are equal, the situation is known as ----------

Select the correct option

- Tie
- Unboundedness
- Degeneracy
- Non-degeneracy

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Question # 2 of 10 (Start time: 10:29:40 PM, 09 February 2022) **Total Marks:** 1

While solving a LP problem by Simplex method in a given iteration, the new basic variable is ---------- and the variable remove from the basis is called ---------.

Select the correct option

○	(leaving, entering)
○	(infeasible, feasible)
○	(feasible, infeasible)
○	(entering, leaving)

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Question # 2 of 10 (Start time: 07:35:04 PM, 09 February 2022)

Total Marks: 1

In M-method, which of the following is true about the coefficient (M) of artificial variable(A) in the objective function?

Select the correct option

- ○ M--› –Infinity
- ○ M--›optimal solution
- ◉ M--›+Infinity
- ○ M--›zero

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Question # 4 of 10 (Start time: 07:36:49 PM, 09 February 2022) **Total Marks:** 1

While solving a Linear Programming problem by Simplex Method, an inequality says:'2x-3y<5' is transformed into strict equality of the form-----, where 's>=0'.

Select the correct option

○	2x-3y -s +A =5
○	2x-3y +s + A=5
○	2x-3y -s =5
○	2x-3y +s =5

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Question # 2 of 10 (Start time: 07:49:56 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be ———— except key row(pivot) entry.

Select the correct option

- ○ strictly negative
- ○ maximum or minimum
- ○ zero ✓
- ○ strictly positive

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MTH601 - Operations Research (Quiz No 2)

Question # 2 of 10 (Start time: 06:58:41 PM, 09 February 2022)

In two phase method, which of the following will be taken as starting solution for 2nd phase?

Select the correct option

- ○ Optimum feasible solution of 1st phase
- ○ Non-basic solution of 1st phase containing non-negative slacks
- ○ Infeasible solution of 1st phase.
- ○ Basic solution of 1st phase containing non-zero artificial variables

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Question # 5 of 10 (Start time: 07:01:39 PM, 09 February 2022) **Total Marks: 1**

While solving a Linear Programming problem by Simplex Method, an inequality says:"2x-3y>5" is transformed into strict equality of the form-----, where "s>=0"

Select the correct option

- 2x-3y -s =5
- 2x-3y +s + A=5
- 2x-3y -s +A =5
- 2x-3y +s =5

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Question # 6 of 10 (Start time: 11:26:05 PM, 09 February 2022) **Total Marks:** 1

In the initial table of Simplex method, the objective function should not contain the terms involving----------variables.

Select the correct option

- basic
- non-basic
- degenerate
- artificial

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Question # 10 of 10 (Start time: 11:34:29 PM, 09 February 2022) **Total Marks: 1**

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

Select the correct option

- (●) all of above ✓
- ( ) Big M
- ( ) simplex
- ( ) Two phase

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MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 10:57:34 PM, 09 February 2022)

A linear programming problem _____ optimal solutions.

Select the correct option

- ○ may have the multiple
- ○ must have the multiple
- ○ must not have the multiple
- ○ may not have the multiple

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Question # 2 of 10 (Start time: 10:57:57 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. Then how many Combinations of Basic feasible Solutions may possible?

Select the correct option

- ○ 2
- ○ 24
- ○ 10
- ○ 15

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MTH601 - Operations Research (Quiz No 2)

Question # 6 of 10 (Start time: 11:11:07 PM, 09 February 2022)

The cost coefficient of artificial variable in Objective function is ----------.

Select the correct option

○	0
○	> than 1
○	1
○	M ✓

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Question # 1 of 10 (Start time: 10:49:30 PM, 09 February 2022)

In an LP problem, to evaluate the basic variables in Simplex method, which of the following method is used?

Select the correct option

- Gauss Elimination method
- Gauss-Jordan method
- Gauss Least Square method
- Gauss-Seidel method

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Question # 1 of 10 (Start time: 10:52:29 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, every entry of objective function row is negative then the given problem ----------.

Select the correct option

- has no solution
- needs further improvement
- has been minimized
- can not be optimized

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Question # 5 of 10 (Start time: 10:01:39 PM, 09 Februar 2022)

In M-method, which of the following is the standard equality corresponds to the constraint:"3x+5y=2"?

Select the correct option

- ○ 3x+2y−A=2
- ◉ 3x+2y−MA=2
- ○ 3x+2y +A=2
- ○ 3x+2y+MA=2

Not confirm

max and min problem are not disused in given statement

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Question # 6 of 10 (Start time: 11:30:41 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem of maximization, if at the end of iteration, every entry of objective function row is positive then the given problem----------.

Select the correct option

- can not be optimized
- has been maximized
- has no solution
- needs further improvement

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Question # 3 of 10 (Start time: 11:03:04 PM, 09 February 2022)

Total Marks: 1

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, there is a positive coefficient in the objective row then the given problem ----------.

Select the correct option

- ○ can not be optimized
- ○ needs further improvement
- ◉ has been maximized
- ○ has no solution

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Question # 2 of 10 (Start time: 10:01:06 PM, 09 February 2022)
Total Marks: 1

While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables.

Select the correct option

- (●) basic
- ( ) non-basic
- ( ) decision
- ( ) artificial

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Question # 5 of 10 (Start time: 09:27:53 PM, 09 February 2022) **Total Marks: 1**

In an LP problem, to evaluate the basic variables in Simplex method in any iteration, we would have to express Objective function in terms of ---------- variables.

Select the correct option

- ○ Slack
- ○ Artificial
- ○ Non-basic
- ◉ Basic

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Question # 3 of 10 (Start time: 10:01:21 PM, 09 February 2022)

Total Marks: 1

If the original Linear Programming problem has no Feasible Solution, then either the Big-M method or Phase-I of the Two-Phase method will give a final solution that has at least one artificial variable _____.

Select the correct option

- less than zero
- of any arbitrary value
- greater than zero
- equal to zero

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Question # 9 of 10 (Start time: 09:52:13 PM, 09 February 2022)

Total Marks: 1

By Simplex method, to minimize 'z = 9x−2y' of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution(0<A<100), which of the following would be the next entering variable?

Select the correct option

- ○ x>0
- ◉ y>0
- ○ y<0
- ○ x<0

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Question # 10 of 10 (Start time: 09:52:20 PM, 09 February 2022)
Total Marks: 1

In M-method, if the given LP problem has the feasible solution then the optimization algorithm enforces artificial variable to--------.

not confirm

Select the correct option

- (●) Zero
- ( ) + Infinity
- ( ) Optimal solution
- ( ) – Infinity

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MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 10:00:55 PM, 09 February 2022)

An optimal solution of a Linear Programming problem is necessarily a Basic feasible solution.

Select the correct option

- ○ False
- ◉ True





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Question # 3 of 10 (Start time: 09:50:58 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

◉	two slack
○	three objective
○	six
○	one artificial

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Question # 6 of 10 (Start time: 09:51:36 PM, 09 February 2022)

Which of the following is true about an LP problem which is solved by M-method and having infeasible solution?

not confirm

Select the correct option

- ○ Associated constraints are not consistent
- ◉ All choices are equivalent. ✓
- ○ Final iteration table include one of the artificial variables.
- ○ Penalty 'M' may not able to force artificial variable to zero in final iteration.

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Question # 8 of 10 (Start time: 09:52:03 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Basic feasible VARIABLES are possible to exists?

Select the correct option

- (●) 4
- ( ) 6
- ( ) 2
- ( ) 10

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MTH601 - Operations Research (Quiz No 2)

Question # 10 of 10 (Start time: 09:45:20 PM, 09 February 2022)

Shortcoming of Big M method is that the value of M could be _____.

Select the correct option

○	negative
○	very small
○	positive
◉	very large





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Question # 1 of 10 (Start time: 09:50:28 PM, 09 February 2022)

Total Marks: 1

In Simplex Standard table to solve an LP problem, if the ratio by taking the RHS of each row and dividing by the corresponding element of the key column is {2/3,5,1/3, -1/2}, then which of the following variable will be leaving?

Select the correct option

- ○ Variable corresponding to '-1/2'
- ○ Variable corresponding to '2/3'
- ◉ Variable corresponding to '1/3' ✓
- ○ Variable corresponding to '5'

not confirm

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Question # 2 of 10 (Start time: 09:50:36 PM, 09 February 2022)

In two phase method process, first phase _____ the sum of artificial variables.

Select the correct option

- (●) minimize
- ( ) non of these
- ( ) maximize or minimize depending on the situation
- ( ) maximize

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Question # 7 of 10 (Start time: 09:44:32 PM, 09 February 2022) **Total Marks: 1**

In the final iteration of M-method to obtain the optimal solution, the artificial variable must------- the basis.

Select the correct option

- subtract from each variable of
- add in each variable of
- include in
- leave ✓

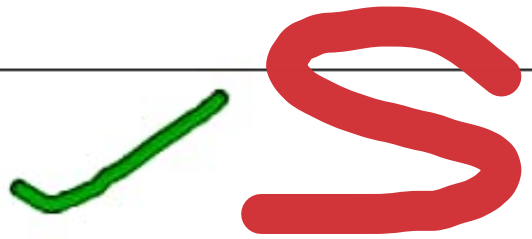
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Question # 9 of 10 (Start time: 09:45:10 PM, 09 February 2022)

When constraints are of greater than or equal to type, then we _____.

Select the correct option

- ○ both of these
- ○ add slack variable and subtract artificial varialbe
- ◉ subtract slack variable and add artificial variable
- ○ non of these



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Question # 10 of 10 (Start time: 09:29:50 PM, 09 February 2022)

The dual of the dual of a Linear Programming problem is itself the _____ problem.

Select the correct option

- ○ dual
- ◉ primal ✓
- ○ transportation
- ○ assignment

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MTH601 - Operations Research (Quiz No 2)

Question # 3 of 10 (Start time: 09:43:31 PM, 09 February 2022)

To remove artificial variable(s) from the objective function, we have _____

Select the correct option

○	simplex
◉	both b and c
○	Big M
○	Two Phase

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Question # 7 of 10 (Start time: 09:29:12 PM, 09 February 2022)

Total Marks: 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, if the initial basic feasible solution is say;(2,4,0,2), then it is ---------- solution.

Select the correct option

- ○ non-degenerate feasible
- ◉ degenerate feasible ✓
- ○ non-degenerate infeasible
- ○ degenerate infeasible

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Question # 8 of 10 (Start time: 09:29:22 PM, 09 February 2022)

Which of the following difficulty may found while attempting an LP problem by M-method?

Select the correct option

○	Degeneracy is inevitable
○	Artificial do not leave the basis
◉	Computational error due to large value of M ✓
○	It often leads to infeasible solution

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MTH601 - Operations Research (Quiz No 2)

Question # 9 of 10 (Start time: 09:29:34 PM, 09 February 2022)

The positive slack variables indicate ----------.

Select the correct option

○	nil resources
○	excess resource available
○	idle(waste) resource
◉	shortage of resources

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Question # 3 of 10 (Start time: 09:27:26 PM, 09 February 2022)

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?

Select the correct option

- (●) Objective function of 1st phase has positive value. 
- ( ) Objective function of 1st phase has zero value.

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Question # 4 of 10 (Start time: 09:27:36 PM, 09 February 2022)

If in a LP problem, the objective is to maximize: z = 2x+3y, with all associated constraints of '=<' type, then in the 1st iteration, z = -

Select the correct option

- ○ 1
- ○ 6
- ○ 5
- ◉ 0 ✓

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Question # 6 of 10 (Start time: 09:28:45 PM, 09 February 2022) **Total Marks: 1**

In two phase method, for the phase-I, if the objective function has zero value with all vanishing artificial variables then we -------.

Select the correct option

- ○ infeasible solution
- ○ cant proceed for 2nd phase
- ○ necessarily have optimal solution
- ◉ proceed for 2nd phase



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Question # 10 of 10 (Start time: 11:31:43 PM, 09 February 2022)

Total Marks: 1

By Simplex method, to maximize 'Z = 2x−9y' of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would be the next entering variable?

Select the correct option

- ○ x<0
- ○ y<0
- ○ y>0
- ◉ x>0 ✓

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MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 09:26:49 PM, 09 February 2022)

In two phase method, for the phase-I, if the given problem has feasible solution then---

Select the correct option

○	objective function is zero but artificial may arbitrary
○	objective function is arbitrary but artificial is zero
◉	both objective function and artificial are zero ✔
○	both objective and artificial can have arbitrary values

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Question # 2 of 10 (Start time: 09:27:16 PM, 09 February 2022) Total Marks: 1

In two phase method if the minimum value of objective function in the first phase is greater than zero, then the solution of original problem _____.

Select the correct option

- non of these
- is uncertain
- exists
- does not exists

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Question # 9 of 10 (Start time: 11:31:30 PM, 09 February 2022)

An unrestricted variable can always be expressed as the difference of two _____ variables.

Select the correct option

- ○ none of these
- ◉ non-negative 
- ○ positive
- ○ negative

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Question # 1 of 10 (Start time: 11:29:46 PM, 09 February 2022)
Total Marks: 1

While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ---------- element.

Select the correct option

- ○ Entering
- ◉ Pivot
- ○ Slack
- ○ Leaving

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MTH601 - Operations Research (Quiz No 2)

Question # 5 of 10 (Start time: 11:27:28 PM, 09 February 2022)

The inequality $3x - 2y \geq 14$ is equivalent to_____.

Select the correct option

○	$3x - 2y \leq 14$
○	$-3x + 2y > 14$
○	$3x - 2y < -14$
◉	$-3x + 2y \leq -14$ 

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MTH601 - Operations Research (Quiz No 2)

Question # 6 of 10 (Start time: 11:27:51 PM, 09 February 2022)

In two phase method, for the phase-I, a new objective function is expressed as -----.

Select the correct option

○	sum of slack variables
○	difference of slack variables
○	difference of artificial variables
◉	sum of artificial variables 

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Question # 7 of 10 (Start time: 11:14:18 PM, 09 February 2022) **Total Marks: 1**

The insensitivity of the solution relative to the original decision variables in an LP problem which is solved by M-method is overcome by ---------.

Select the correct option

- ○ Duality principle
- ○ Graphical method
- ◉ Two phase method ✓
- ○ Simplex method

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Question # 10 of 10 (Start time: 11:14:47 PM, 09 February 2022)

Total Marks: 1

In a feasible region, if the end points of the line segment are basic solutions, then all the points between these two will ----------.

Select the correct option

- be non basic
- be optimal
- be infeasible
- also be the basic

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Question # 2 of 10 (Start time: 11:17:19 PM, 09 February 2022) **Total Marks: 1**

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

Select the correct option

- (2,2)
- (1,2)
- (-4,-2)
- (0,0) ✓

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sec(s)

MTH601 - Operations Research (Quiz No 2)

Quiz Start Time: 11:06 PM, 09 February 2022

Question # 6 of 10 (Start time: 11:07:47 PM, 09 February 2022)

Total Marks: 1

If an artificial variable is not included while converting a constraint of type ">=" into equation, then we will have -------- solution.

Select the correct option

- degenerate
- non-degenerate
- feasible
- infeasible 

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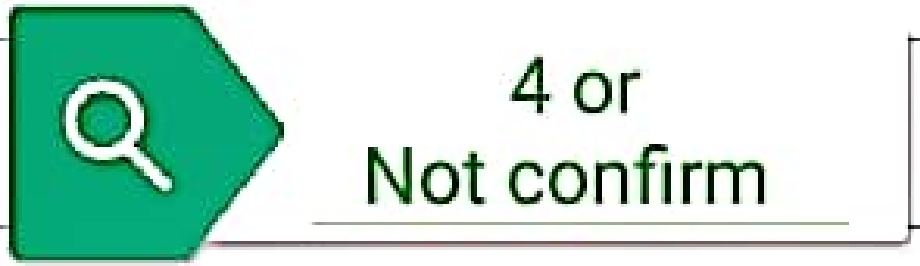
Question # 9 of 10 (Start time: 11:08:45 PM, 09 February 2022)

Total Marks: 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Infeasible VARIABLES are possible to exists?

Select the correct option

- (●) 10
- ( ) 2
- ( ) 4
- ( ) 6



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MTH601 - Operations Research (Quiz No 2)

Question # 1 of 10 (Start time: 11:12:43 PM, 09 February 2022)

When using Big M method, the objective function should not contain _____ variable(s).

Select the correct option

○	artificial and slack
◉	artificial
○	x_1
○	slack

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Question # 2 of 10 (Start time: 11:06:47 PM, 09 February 2022) Total Marks: 1

In Big M-method, if the objective function is expressed in terms of non-basic variable then the coefficient of artificial variable in this iteration must be --------.

Select the correct option

- M
- zero ✓
- -M
- one

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Question # 3 of 10 (Start time: 11:06:57 PM, 09 February 2022) Total Marks: 1

In two phase method, for the phase-I, the problem has infeasible solution if the minimum value of objective function ----------.

Select the correct option

- (●) greater than zero
- ( ) less than zero
- ( ) is imaginary
- ( ) equal to zero

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Question # 5 of 10 (Start time: 11:07:34 PM, 09 February 2022) **Total Marks: 1**

If initial basic solution is ----------, while solving an LP problem then no further iteration can be performed.

Select the correct option

- (●) infeasible
- ( ) degenerate
- ( ) feasible
- ( ) non-degenerate

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MTH601 - Operations Research (Quiz No 2) Quiz Start Time: 11:06 PM, 09 February 2022

Question # 1 of 10 (Start time: 11:06:39 PM, 09 February 2022) Total Marks: 1

Which of the following order pair would minimize the objective function of the linear programming problem; $z = x + 5y$ subject to $x \geq 2, y \geq 0$?

Select the correct option

Reload Math Equations

○	(2,0)
○	(2,3)
◉	(0,0)
○	(0,3)

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Question # 2 of 10 (Start time: 11:02:47 PM, 09 February 2022)

In Simplex method, which of the following is the standard equality corresponding to the constraint"3x+5y>=2"?

Select the correct option

◉	3x+2y−S+A=2 
○	3x+2y+S+A=2
○	3x+2y−S−A=2
○	3x+2y+S−A=2

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Question # 1 of 10 (Start time: 11:02:35 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

○	one slack
◉	two artificial 
○	six
○	three objective

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Question # 10 of 10 (start time: 10:56:14 PM, 09 February 2022) **Total Marks: 1**

Which of the following is the standard form of objective function corresponding to,"Minz=2x−11y, subject to x=5 and y=7"? Where A's are artificial.

Select the correct option

- z = 2x − 11y −MA1 − MA2
- z = 2x − 11y +MA1 − MA2
- z = 2x − 11y +MA1 + MA2 ✓
- z = 2x − 11y −MA1 + MA2

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Question # 7 of 10 (Start time: 10:09:34 PM, 09 February 2022) **Total Marks: 1**

In the Simplex method to solve an LP problem of maximization, if at the end of an iteration, there is a negative coefficient in the objective row then the given problem---------.

Select the correct option

- ○ has been maximized
- ◉ needs further improvement 
- ○ cant be optimized
- ○ has no solution

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Question # 6 of 10 (Start time: 10:55:13 PM, 09 February 2022)

For $Z = 4x_1 + x_2$ subject to $3x_1 + 4x_2 \geq 20, x_1 + 5x_2 \geq 15$, we have _____ variables.

Select the correct option

○	three objective
◉	both a and b
○	two slack
○	two artificial

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Question # 9 of 10 (Start time: 10:56:05 PM, 09 February 2022) Total Marks: 1

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

Select the correct option

- ○ may have a solution
- ○ non of these
- ○ must have a solution
- ◉ does not have a solution 

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Question # 7 of 10 (Start time: 10:05:36 PM, 09 February 2022)

In two phase method, for the phase-I, a new objective function in terms of artificial is to be ----------.

Select the correct option

○	maximized
◉	minimized

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MTH601 - Operations Research (Quiz No 2)

Question # 8 of 10 (Start time: 10:05:52 PM, 09 February 2022)

A Linear Programming problem may not have any feasible solution.

Select the correct option

◉	True
○	False

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Question # 9 of 10 (Start time: 10:06:05 PM, 09 February 2022)

Which of the following is true about the inclusion of non-negative slack variable into a constraint of type 'less than or equal'?

Select the correct option

○	This inclusion doest not affect the inequality
○	Inequality become redundant and hence skips from the LP problem.
○	Inequality is transformed into type: 'greater than or equal'
◉	Inequality is transformed into strict equality constraint 

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sec(s)

MTH601 - Operations Research (Quiz No 2)

Quiz Start Time: 10:08 PM, 09 February 2022

Question # 6 of 10 (Start time: 10:09:25 PM, 09 February 2022)

Total Marks: 1

In two phase method if the minimum value of objective function in the first phase is zero, then the solution of original problem _____.

Select the correct option

- non of these
- is uncertain
- does not exists
- exists 

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Question # 2 of 10 (Start time: 10:04:36 PM, 09 February 2022) **Total Marks: 1**

While solving an LP problem by Simplex method, if we have '3' constraint equations in '5' variables then to find basic variables we have to put any ------- variables equal to zero.

Select the correct option

- ○ 4
- ○ 2
- ◉ 3
- ○ 1

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MTH601 - Operations Research (Quiz No 2)

Question # 3 of 10 (Start time: 10:04:51 PM, 09 February 2022)

In graphical solution of solving Linear Programming problem to convert inequalities into equations, we ----------.

Select the correct option

○	use Artificial surplus variables
◉	simply assume them to be equations 
○	use Surplus variables
○	use Slack variables

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MTH601 - Operations Research (Quiz No 2)

Question # 4 of 10 (Start time: 10:05:01 PM, 09 February 2022)

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Select the correct option

◉	Basis 
○	Non-basic variables' set

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Time Left sec(s)

MTH601 - Operations Research (Quiz No 2)

Quiz Start Time: 10:00 PM, 09 February 2022

Question # 7 of 10 (Start time: 10:02:17 PM, 09 February 2022)

Total Marks: 1

In a feasible region, if the end points of a line segment are basic solutions which BOTH also give the optimal solution, then ---------- will also serve for optimal.

Select the correct option

- (●) all other points on the segment
- ( ) the point which divides segment in ratio 1:2
- ( ) no other point
- ( ) the only midpoint of the segment

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1. **When money value changes with time at 10 %, then PWF for the first year is**

(a) 1

(b) 0.909

(c) 0.852

(d) 0.9

Answer: Option ***(a)***

2. **When money value changes with time at 20%, the discount factor for the 2nd year is**

(a) 1

(b) 0.833

(c) 0

(d) 0.6955

Answer: Option **(b)**

3. **Which of the following replacement policies is considered to be dynamic in nature?**

(a) Time is a continuous variable and the money value does not change with time

(b) When money value does not change with time and time is a discrete variable

(c) When money value changes with time

(d) When money value remains constant for some time and then goes on changing with time

Answer: Option **(c)**

4. **When the probability of failure reduces gradually, the failure mode is said to be**

(a) Regressive

(b) Retrogressive

(c) Progressive

(d) Recursive

Answer: Option **(b)**

5. **..... is concerned with the determination of the most economic *replacement policy*.**

(a) Probabilistic programming

(b) *Linear programming*

(c) Search theory

(d) Replacement theory

Answer: Option **(d)**

6. **Which cost of the following is irrelevant to replacement analysis?**

(a) Purchase cost of the machine

(b) Operating cost of the machine

(c) Maintenance cost of the machine

(d) Machine hour rate of the machine

Answer: Option **(d)**

7. **Group replacement policy is most suitable for**

(a) Trucks

(b) Infant machines

(c) Street light bulbs

(d) New cars

Answer: *Option* **(c)**

8. **Replacement of an item will become necessary when**

(a) an old item becomes too expensive to operate or maintain

(b) when your operator desires to work on a new machine

(c) when your opponent changes his machine in his unit

(d) when the company has surplus funds to spend

Answer: Option **(a)**

9. **The following is not discussed in group replacement policy**

(a) Failure Probability

(b) Cost of individual replacement

(c) Loss due to failure

(d) Present worth factor series

Answer: Option **(d)**

10. **It is assumed that maintenance cost mostly depends on**

(a) calendar age

(b) manufacturing date

(c) running age

(d) user's age

Answer: Option **(c)**

11. Replacement decision is very much common in this stage

(a) Infant stage

(b) Old age

(c) Youth

(d) In all the above

Answer: Option **(b)**

12. The replacement policy that is imposed on an item irrespective of its failure is

(a) Group replacement

(b) Individual replacement

(c) Repair spare replacement

(d) Successive replacement

Answer: Option **(a)**

13. The total number of allocations in a basic feasible solution to the transportation problem of $m \times n$ size is equal to

(a) $m \times n$

(b) $\left(\frac{m}{n}\right) - 1$

(c) $m + n + 1$

(d) $m + n - 1$

Answer: Option **(d)**

14. Contractual maintenance or agreement maintenance with the manufacturer is suitable for equipment, which is

(a) In its infant state

(b) When a *machine is an old one*

(c) Scrapped

(d) None of these

Answer: Option **(a)**

15. **Which of the following maintenance policy is not used in the old age stage of a machine**

(a) Operate up to failure and do corrective maintenance

(b) Reconditioning

(c) Replacement

(d) Scheduled preventive maintenance

Answer: Option **(d)**

16. **The following replacement model is said to be a probabilistic model**

(a) When money value does not change with time and time is a continuous variable

(b) When money value changes with time

(c) When money value does not change with time and time is a discrete variable

(d) Preventive maintenance policy

Answer: Option **(d)**

17. **A machine is replaced with an average running cost**

(a) Is not equal to the current running cost

(b) till the current period is greater than that of next period

(c) of the current period is greater than that of the next period,

(d) of the current period is less than that of next period

Answer: Option **(d)**

18. **Decreasing failure rate is usually observed in ..... stage of the machine**

(a) infant

(b) youth

(c) old age

(d) any time in its life

Answer: Option **(a)**

19. **The type of failure that usually occurs in old age of the machine is**

(a) Random failure

(b) Early failure

(c) Chance failure

(d) Wear - out failure

Answer: Option **(d)**

20. **The production manager will not recommend group replacement policy in *case of***

(a) When large number of identical items is to be replaced

(b) Low cost items are to be replaced, where record keeping is a problem

(c) For items that fail completely

(d) Repairable items

Answer: Option **(d)**

21. In replacement analysis the maintenance cost is a function of

(a) time

(b) resale value

(c) initial investment

(d) none of these

Answer: Option **(a)**

22. Which of the following is the correct assumption for replacement policy when money value does not change with time?

(a) No Capital cost

(b) No scrap value

(c) Constant scrap value

(d) Zero maintenance cost

Answer: Option **(c)**

23. Which one of the following does not match the group?

(a) Present Worth Factor (PWF)

(b) Discounted rate (DR)

(c) Depreciation value (DV)

(d) Mortality Tables (MT)

Answer: Option **(d)**



24. Group replacement policy applies to

(a) Irreparable items

(b) Repairable items

(c) Items that fail partially

(d) Items that fail completely & suddenly

Answer: Option **(d)**

25. *If a machine becomes old, then the failure rate expected will be*

(a) Constant

(b) Increasing

(c) decreasing

(d) we cannot say

Answer: Option **(b)**

26. Replacement is said to be necessary if

(a) Failure rate is increasing

(b) Failure cost is increasing

(c) Failure probability is increasing

(d) Any of these

Answer: Option **(d)**

27. **In this stage, the machine operates at highest efficiency and its production rate will be high**

(a) Infant stage

(b) Youth stage

(c) Old age

(d) None of these

Answer: Option **(b)**

28. **In retrogressive failures, the failure probability ..... with time**

(a) increases

(b) remains constant

(c) decreases

(d) none of these

Answer: Option **(c)**

MTH601 By Sparrow

Final Term

Fall 2021-2022

Sparrows

1. **When money value changes with time at 10 %, then PWF for the first year is**
   - **A. 1**
   - B. 0.909
   - C. 0.852
   - D. 0.9

2. **When money value changes with time at 20%, the discount factor for the 2nd year is**
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   - C. 0
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   - A. Time is a continuous variable and the money value does not change with time
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   - A. Regressive
   - **B. Retrogressive**
   - C. Progressive
   - D. Recursive

5. **_____ is concerned with the determination of the most economic replacement policy.**
   - A. Probabilistic programming

B. Linear programming
C. Search theory
D. Replacement theory

6. Which cost of the following is irrelevant to replacement analysis?
A. Purchase cost of the machine
B. Operating cost of the machine
C. Maintenance cost of the machine
D. Machine hour rate of the machine

7. If a balanced Transportation problem with '7' sources and '6'sinks then its non-degenerate basic feasible solution will have---------- independent allocations.
A. 10

8. Group replacement policy is most suitable for
A. Trucks
B. Infant machines
C. Street light bulbs
D. New cars

9. Replacement of an item will become necessary when
A. an old item becomes too expensive to operate or maintain
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10. The following is not discussed in group replacement policy
A. Failure Probability
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A. Infant stage

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B. Old age
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A. M+N – 1

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B. When a machine is an old one
C. Scrapped
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C. Replacement
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A. When money value does not change with time and time is a continuous variable
B. When money value changes with time
C. When money value does not change with time and time is a discrete variable
D. Preventive maintenance policy

18. A machine is replaced with an average running cost

A. Is not equal to the current running cost
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C. of the current period is less than that of next period

19. Decreasing failure rate is usually observed in _____ stage of the machine

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A. infant
B. youth
C. old age
D. any time in its life

20. The type of failure that usually occurs in old age of the machine is
A. Random failure
B. Early failure
C. Chance failure
D. Wear - out failure

21. For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?
A. 120

22. The production manager will not recommend group replacement policy in case of
A. When large number of identical items is to be replaced
B. Low cost items are to be replaced, where record keeping is a problem
C. For items that fail completely
D. Repairable items

23. In replacement analysis the maintenance cost is a function of
A. time
B. resale value
C. initial investment
D. none of these

24. Which of the following is the correct assumption for replacement policy when money value does not change with time?
A. No Capital cost
B. No scrap value
C. Constant scrap value
D. Zero maintenance cost

25. In the assignment problem, the cost matrix is _____.
E. rectangular

26. Which one of the following does not match the group?

Sparrows

A. Present Worth Factor (PWF)
B. Discounted rate (DR)
C. Depreciation value (DV)
D. Mortality Tables (MT)

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B. Repairable items
C. Items that fail partially
D. Items that fail completely & suddenly

28. If a machine becomes old, then the failure rate expected will be
A. Constant
B. Increasing
C. decreasing
D. we cannot say

29. Replacement is said to be necessary if
A. Failure rate is increasing
B. Failure cost is increasing
C. Failure probability is increasing
D. Any of these

30. In this stage, the machine operates at highest efficiency and its production rate will be high
A. Infant stage
B. Youth stage
C. Old age
D. None of these

31. In retrogressive failures, the failure probability _____ with time
A. increases
B. remains constant
C. decreases
D. none of these

32. The objective of a transportation problem is to develop an _____ transportation schedule that meets all demand from given stock at a _____ total shipping cost.
A. (integral, maximum)

33. For North West Corner method, in the first row and first column, resource and sink

Sparrows

contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

A. (2,2)

34. While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ------ element.

A. Pivot

35. Search and cutting methods are used to solve following types of problems

A. Transportation
B. Assignment
B. Integer Programming
C. Queuing problems

36. To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce

A. Parallel Activity
B. Dummy Activity
C. Sequential Activity
D. None of these

37. The positive slack variables indicate _____.

A. excess resource available (sure)

38. The variable is chosen by examining the cost coefficients in the objective function.

A. Entering
B. Leaving

39. In retrogressive failures, the failure probability _____ with time

A. Increases
B. Remains
C. Constant
D. Decreases

40. Decreasing failure rate is usually observed in_____ stage of the machine

A. Infant
B. Youth

41. When money value changes with time at 10 %, then PWF for first year is

A. 1

Sparrows

B. 0.909
C. 0.852

42. The rent for the stores where materials are stored falls under

A. Inventory carrying cost
B. Ordering cost

43. O.R. Society of India is founded in the year

A. 1965
B. 1970
C. 1959

44. The rule which determines the decision at each stage is known as

A. State
B. Stage
C. Policy
D. Decision

45. In dynamic programming problems, the main problem is divided into sub problems. Each sub problem is known as

A. Part
B. Stage
C. State

46. A competitive situation is known as

A. Competition
B. Marketing
C. Game
D. None of the above

47. In case, there is no saddle point in a game then the game is

A. Deterministic game
B. Fair game
C. Mixed strategy game
D. Multi player game

48. When Minimax and Maximum criteria matches, then

A. Fair game is exists
B. Unfair game is exists
C. Mixed strategy exists
D. Saddle point exists

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49. If the value of the game is zero, then the game is known as

A. Fair strategy
B. Pure strategy
C. Pure game
D. Mixed strategy

50. The objective of network analysis is to_____.

A. minimize total project duration
B. minimize total project cost
C. minimize production delays, interruption and conflicts
D. maximize total project duration

51. Service mechanism in a queuing system is characterized by _____.

A. customers behavior
B. servers behavior
C. customers in the system
D. server in the system

52. The difference between total float and head event slack is _____

A. free float
B. independent float
C. interference float
D. linear float

53. For any primal problem and its dual_____.

A. optimal value of objective function is same
B. dual will have an optimal solution iff primal does too
C. primal will have an optimal solution iff dual does too
D. both primal and dual cannot be infeasible

54. If any value in XB column of final simplex table is negative, then the solution is _____.

A. Infeasible
B. Infeasible
C. Bounded
D. no solution

55. Operations Research approach is _____.

A. multi-disciplinary

Sparrows

B. scientific
C. intuitive
D. collect essential data

56. If there is no non-negative replacement ratio in solving a Linear Programming Problem then the solution is _____.

A. Feasible
B. Bounded
C. Unbounded
D. Infinite

57. The calling population is considered to be infinite when _____.

A. all customers arrive at once
B. capacity of the system is infinite
C. service rate is faster than arrival rate
D. arrivals are independent of each other

58. In marking assignments, which of the following should be preferred?

A. Only row having single zero
B. Only column having single zero
C. Only row/column having single zero
D. Column having more than one zero

59. A petrol pump has one pump; Vehicles arrive at the petrol pump according to poison input process at average of 12 per hour. The service time follows exponential distribution with a mean of 4 minutes. The pumps are expected to be idle for _____.

A. 3/5
B. 4/5
C. 5/3
D. 6/5

60. If the order quantity (size of order) is increased, _____

A. holding costs decrease and ordering costs increase
B. holding costs increase and ordering costs decrease
C. the total costs increase and then decrease
D. storage cost as well as stock-out cost increase

61. A mixed strategy game can be solved by _____.

A. Simplex method

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B. Hungarian method
C. Graphical method
D. Degeneracy

62. The activity cost corresponding to the crash time is called the _____.
A. critical time
B. normal time
C. cost slope
D. crash cost

63. A set of feasible solution to a Linear Programming Problem is _____
A. convex
B. polygon
C. triangle
D. bold

64. If the primal problem has n constraints and m variables then the number of constraints in the dual problem is _____.
A. mn
B. m+n

65. The non-basic variables are called _____.
A. shadow cost
B. opportunity cost
C. slack variable
D. surplus variable

66. Key element is also known as _____.
A. slack
B. surplus
C. artificial
D. pivot

67. The allocation cells in the transportation table will be called _____ cell
A. Occupied

68. **To resolve degeneracy at the initial solution, a very small quantity is allocated in _____ cell**
A. occupied
B. unoccupied

Sparrows

69. An assignment problem is a particular case of _____.

A. transportation Problem
B. assignment Problem
C. travelling salesman problem
D. replacement Problem

70. Using _____ method, we can never have an unbounded solution

A. Simplex
B. Dual simplex

71. The customers of high priority are given service over the low priority customers is _____.

A. Pre-Emptive

72. A queuing system is said to be a _____ when its operating characteristic are independent upon time

A. pure birth model
B. pure death model
C. transient state
D. steady state

73. The difference between total and free float is _____.

A. total
B. free
C. independent
D. interference

74. The number of time estimates involved in Program Evaluation Review Technique problem is _____.

A. 1
B. 2
C. 3
D. 4

75. The assignment problem is always a _____matrix.

A. circle
B. square
C. rectangle
D. triangle

Sparrows

76. The slack variables indicate _____.

A. excess resource available.
B. shortage of resource
C. nil resource
D. idle resource

77. While solving a linear programming problem infeasibility may be removed by _____.

A. adding another constraint
B. adding another variable
C. removing a constraint
D. removing a variable

78. The right hand side constant of a constraint in a primal problem appears in the corresponding dual as_____.

A. a coefficient in the objective function

79. During iteration while moving from one solution to the next, degeneracy may occur when_____

A. the closed path indicates a diagonal move
B. Two or more occupied cells on the closed path with minus sign are tied for lowest circled value.

80. Maximization assignment problem is transformed into a minimization problem by_____.

A. adding each entry in a column from the maximum value in that column
B. subtracting each entry in a column from the maximum value in that column
C. subtracting each entry in the table from the maximum value in that table
D. adding each entry in the table from the maximum value in that table

81. Priority queue discipline may be classified as_____.

A. pre-emptive or non-pre-emptive
B. limited
C. unlimited
D. finite

82. Replace an item when_____.

A. average cost upto date is equal to the current maintenance cost

Sparrows

B. average cost upto date is greater than the current maintenance cost
C. Average cost upto date is less than the current maintenance cost.
D. next year running cost in more than average cost of nth year

83. In time cost trade off function analysis_____.

A. cost decreases linearly as time increases
B. cost increases linearly as time decreases
C. cost at normal time is zero
D. cost increases linearly as time increases

84. The minimum number of lines covering all zeros in a reduced cost matrix of order n can be _____.

A. at the most n
B. at the least n
C. n-1
D. n+1

85. For a 2.5% increase in order quantity (under fundamental EOQ problem) the total relevant cost would _____

A. Increase by 2.5%.
B. Decrease by 2.5%.

86. In the basic EOQ model, if the lead time increases from 2 to 4 days, the EOQ will _____

A. double increase
B. remain constant
C. but not double
D. decrease by a factor of two

87. In the network, one activity may connect any _____ nodes

A. 1
B. 2
C. 3
D. 4

88. Graphical method is also known as _____.

A. Simplex Method
B. Dual Simplex Method
C. Big-M Method
D. Search-Approach Method

Sparrows

89. Linear Programming Problem is a technique of finding the _____.

A. optimal value
B. approximate value
C. initial value
D. infeasible value

90. The irreducible minimum duration of the project is called _____.

A. critical time
B. normal time
C. cost slope
D. crash duration

91. The cost of a slack variable is _____.

A. 0
B. 1
C. 2
D. -1

92. The server utilization factor is also known as _____

A. exponential distribution
B. traffic intensity

93. A Linear Programming Problem have _____ optimal solution

A. 1
B. 2
C. more than 1

94. The models in which only arrivals are counted and no departure takes place are called _____.

A. pure birth model
B. pure death mode

95. _____ of a queuing system is the state where the probability of the number of customers in the system depends upon time

A. pure birth model
B. pure death model
C. transient state
D. steady state

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96. The initial event which has all outgoing arrows with no incoming arrow is numbered _____.

A. 0
B. 1
C. -1
D. 2

97. In a network diagram an event is denoted by the symbol _____.

A. arrow
B. straight line
C. curve
D. circle

98. _____ is used for non-repetitive jobs

A. Queue
B. Replacement
C. CPM
D. PERT

99. An _____ represent the start or completion of some activity and as such it consumes no time

A. activity
B. event
C. slack
D. path

100. The region common to all the constraints including the non-negativity restrictions is called the _____.

A. solution space
B. unique solution
C. optimum solution
D. infeasible solution

101. Operation research approach is typically based on the use of _____.

A. Physical model.
B. Mathematical model.
C. Iconic model.
D. Descriptive model.

102. For a salesman who has to visit n cities, following are the ways of his tour plan _____.

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A. n!
B. (n+a)!
C. (n-a)!
D. N

103. The problem of replacement is felt when job performing units fail _____.

A. suddenly and gradually
B. gradually
C. suddenly
D. neither gradually nor suddenly

104. Float analysis is useful for _____.

A. total float
B. free float
C. independent float
D. variance of each float

105. The difference between free float and tail event slack is _____

A. total float
B. independent float
C. interference float
D. slack

106. The average arrival rate in a single server queuing system is 10 customers per hour and average service rate is 15 customers per hour. The average time that a customer must wait before it is taken up for service shall be _____minutes.

A. 6
B. 8
C. 10
D. 12

107. The time between the placement of an order and its delivery is called as _____

A. buffer time
B. lead time
C. Economic Order Quantity
D. capital time

Sparrows

108. All the basis for a transportation problem is _____.

A. square
B. rectangle
C. diagonal
D. triangle

109. The linear function to be maximized or minimized is called _____ function.

A. injective
B. surjective
C. bijective
D. optimal

110. The coefficient of an artificial variable in the objective function of penalty method is always assumed to be _____.

A. 0
B. 1
C. M
D. -M

111. The process that performs the services to the customer is known as _____.

A. queue
B. service channel
C. customers
D. server

112. A queuing system is said to be a _____ when its operating characteristic are dependent upon time

A. pure birth model
B. pure death model
C. transient state
D. steady state

113. Slack is also known as _____.

A. Float
B. event
C. activity

Sparrows

D. path

114. What type of distribution does a time follow in program evaluation review technique model?

A. Poisson
B. Exponential
C. Normal
D. Chi Square

115. A activity in a network diagram is said to be _____ if the delay in its start will further delay the project completion time.

A. Critical
B. critical path
C. crash
D. non critical

116. In the optimal simplex table, Zj-Cj=0 value indicates _____.

A. alternative solution
B. bounded solution
C. infeasible solution
D. unbounded solution

117. In an assignment problem involving 5 workers and 5 jobs, total number of assignments possible are _____.

A. 5
B. 10
C. 15
D. 20

118. Average number of trains spent in the yard is denoted by _____.

A. E(n)
B. E(m)
C. E(v)
D. E(w)

119. The dual of the dual is _____.

A. dual-primal
B. primal-dual
C. dual
D. primal

Sparrows

120. All equality constraints can be replaced equivalently by _____ inequalities

A. 1
B. 2
C. 3
D. 4

121. If the primal has an unbound objective function value then the dual has _____.

A. solution
B. basic solution
C. basic feasible solution
D. no feasible solution

122. An activity is represented by a/an _____.

A. arrow
B. straight line
C. curve
D. arc

123. The penalty for not taking correct decision is known as _____.

A. fine
B. loss
C. cost
D. opportunity cost

124. Every basic feasible solution of a general assignment problem having a square pay-off matrix of order n should have assignments equal to_____.

A. 2n-1
B. n
C. n+1
D. n-2

125. A game is said to be fair if _____.

A. lower and upper values are zero
B. only lower value to be zero
C. only upper value to be zero
D. lower and upper values are not equal to zero

Sparrows

126. When D=18000, holding cost=Rs.1.20, set-up cost=Rs.400 ,EOQ = _____

A. 3465
B. 3750
C. 3500
D. 4000

127. Given arrival rate = 15/hr, service rate = 20/hr, the value of traffic intensity is _____.

A. 3/4
B. 4/3
C. 3/5
D. 4/5

128. A solution can be extracted from a model either by

A. Conducting experiments on it
B. Mathematical analysis
C. Both A and B
D. Diversified Techniques

129. An assignment problem can be viewed as a special case of transportation problem in which the capacity from each source is _____ and the demand at each destination is _____.

A. 1; 1
B. Infinity; infinity
C. 0; 0
D. 1000; 1000
E. -1; -1

130. Hungarian Method is used to solve

A. A transportation problem
B. A travelling salesman problem

131. In Degenerate solution value of objective function _____.

A. One or more basic variables are zero

132. In game theory, the outcome or consequence of a strategy is referred to as the

Sparrows

A. Payoff.
B. Penalty.

133. Operations research is the application of _____ methods to arrive at the optimal Solutions to the problems.
A. economical
B. scientific
C. a and b both

134. The degeneracy in the transportation problem indicates that
A. Dummy allocation(s) needs to be added
B. The problem has no feasible solution
C. The multiple optimal solution exist

135. The initial solution of a transportation problem can be obtained by applying any known method. However, the only condition is that
A. The solution be optimal
B. The rim conditions are satisfied
C. The solution not be degenerate
D. All of the above

136. The method of finding an initial solution based upon opportunity costs is called _____.
A. the northwest corner rule
B. Vogel's approximation
C. Johansson's theorem

137. The objective function and constraints are functions of two types of variables, _____ variables and _____ variables.
A. Positive and negative
B. Controllable and uncontrollable

138. The objective function for a minimization problem is given by
z = 2 x1 - 5 x2 + 3 x3
The hyperplane for the objective function cuts a bounded feasible region in the space
(X1, x2, and x3). Find the direction vector d, where a finite optimal solution can be reached.
A. d(2,-5,3)

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B. d(-2,5,-3)

C. d(2,5,3)

139. What have been constructed from OR problems and methods for solving the models that are available in many cases?

A. Scientific Models

B. Algorithms

C. Mathematical Models

Sparrows

Contact for Help.

We have checked all the Answers if you find any mistake let us know.

Sparrow +923313000225.

Remember in Prayers.

Best Of Luck.

Creative Ahmad Pk

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

Select the correct option

- ○ False
- ● True ✓

Creative Ahmad Pk

Question # 7 of 10 (Start time: 09:19:56 PM, 08 March 2022) Total Marks: 1

During the substitution process of determining the Transportation Cost 'xij' in a balanced Transportation problem, no--------- operation is used as the coefficients of 'xij' are '0' or '1'. Due to this property, every basic feasible solution have only non negative integral values

Select the correct option

- ○ Division
- ● Addition ✓ N.C

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Question # 3 of 10 (Start time: 08:56:41 PM, 08 March 2022) Total Marks: 1

If there is infinite number of servers then all the customers are served ---------- on arrival.

Select the correct option

- ○ randomly
- ● instantaneously ✓

Creative Ahmad Pk

In assignment problems, the tasks are assigned to the individuals in such a way that the total time to do a work is _____.

Select the correct option

- ○ maximized
- ● minimized ✓
- ○ minimized or maximized
- ○ all of above

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If a Transportation problem has 'm+n' constraints then how many number of unknowns it will have?

Select the correct option

- ○ m/n
- ○ m.n
- ○ m−n
- ● m+n ✓

Creative Ahmad Pk

Question # 5 of 10 (Start time: 09:18:00 PM, 08 March 2022) Total Marks:

The characteristics of a queuing model is independent of----------.

Select the correct option

- ● Limit of length of queue ✓
- ○ Service Pattern
- ○ Queue discipline
- ○ Number of service stations

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Question # 10 of 10 (Start time: 09:24:22 PM, 08 March 2022) Total Marks: 1

If a balanced Transportation problem with '7' sources and '6'sinks then its non-degenerate basic feasible solution will have---------- independent allocations.

Select the correct option

- [x] 10
- [ ] 11
- [ ] 13
- [ ] 12

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Question # 9 of 10 (Start time: 09:23:13 PM, 08 March 2022) Total Marks: 1

Hungarian Algorithm is used to solve _____problem.

Select the correct option

- [ ] transportation
- [ ] both a and b
- [x] assignment
- [ ] none of these

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Question # 8 of 10 (Start time: 09:20:59 PM, 08 March 2022) Total Marks:

Distribution of service time in Queuing Model is ---------- exponential distribution.

Select the correct option

- ○ divided
- ○ negative
- ○ multiplicative
- ● positive ✓

Creative Ahmad Pk

Question # 6 of 10 (Start time: 09:28:49 PM, 08 March 2022) Total Marks: 1

During the substitution process of determining the Transportation Cost 'xij' in a balanced Transportation problem, no--------- operation is used as the coefficients of 'xij' are '0' or '1'. Due to this

Select the correct option

- ✓ Division
- ○ Addition

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Question # 5 of 10 (Start time: 09:27:44 PM, 08 March 2022) Total Marks:

In the assignment problem, the cost matrix is ----------.

Select the correct option

- singular
- square
- non-singular
- rectangular ✓

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Question # 4 of 10 (Start time: 09:26:49 PM, 08 March 2022) Total M

The arrival rate of customers in a Queue is an example of ---------- variable.

Select the correct option

- stochastic ✓
- discrete
- continuous
- complex

Creative Ahmad Pk

Question # 8 of 10 (Start time: 09:30:12 PM, 08 March 2022) Total Marks: 1

Which of the following statement applies to both transportation model and assignment model?

Select the correct option

- [x] Both have objective function, structural constraint and non-negativity constraints
- [ ] Both use Vogel's approximation for getting basic feasible solution
- [ ] Both are tested by Steping Stone method for optimality
- [ ] The inequalities of both problems are related to one type of resource.

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The assignment problem is unbalanced if the cost matrix is not a ----------matrix.

Select the correct option

- [ ] rectangular
- [ ] singular
- [ ] non-singular
- [x] square

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Question # 9 of 10 (Start time: 09:31:58 PM, 08 March 2022)

Distribution of service time in Queuing Mode
-------exponential distribution.

Select the correct option

- multiplicative
- negative
- positive
- divided

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MTH601:Quiz No 3 2022

Question # 3 of 10 (Start time: 09:40:12 PM, 08 March 2022) Total Marks: 1

The row, which is introduced in the matrix to balance an unbalnced Transportation problem, is known as ----------.

Select the correct option

- dummy row
- slack row
- key row
- idle row

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Question # 4 of 10 (Start time: 09:41:17 PM, 08 March 2022) To

Hungarian Algorithm is used to solve _____problem.

Select the correct option

- transportation
- assignment
- both a and b
- none of these

Creative Ahmad Pk

Which of the following department is more responsible for the development of queuing theory?

elect the correct option

- Telephone department
- Municipal office
- Railway station
- Health department

Creative Ahmad Pk

Question # 5 of 10 (Start time: 09:42:05 PM, 08 March 2022) Total Marks: 1

Distribution of service time in Queuing Model is ----------exponential distribution.

Select the correct option

- negative
- multiplicative
- positive
- divided

Creative Ahmad Pk

In a queue, the arrival pattern can be expressed in terms of ---------.

Select the correct option

- number of servers
- customer behavior
- Probabilities
- exact continuous variables

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Question # 8 of 10 (Start time: 09:45:40 PM, 08 March 2022) Total Ma

To balance Assignment matrix we have to----------.

Select the correct option

- open a Dummy column
- open either a dummy row or column depending on the situation
- open a Dummy row
- you cannot balance the assignment matrix

Creative Ahmad Pk

Question # 7 of 10 (Start time: 09:44:40 PM, 08 March 2022) Total Marks:

If there is infinite number of servers then all the customers are served ---------- on arrival.

Select the correct option

- randomly
- instantaneously

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MTH601:Quiz No 3

Question # 2 of 10 (Start time: 09:51:11 PM, 08 March 2022)

In any Replacement Model a contractual maintenance or agreement maintenance with manufacturer is suitable for equipment, which is ----------

Select the correct option

- None of the above
- when machine is old one
- in its infant state
- scrapped

Processing request...

Creative Ahmad Pk

Question # 3 of 10 (Start time: 09:52:23 PM, 08 March 2022)

Which of the following cost is irrelevant to replacement analysis?

Select the correct option

- Purchase cost of the machine
- Operating cost of the machine
- Maintenance cost of the machine
- Machine hour rate of the machine

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Question # 1 of 10 (Start time: 09:50:06 PM, 08 March 2022)

The configuration of an assignment problem is similar to that of --------- problem.

Select the correct option

- [x] Transportation
- [ ] Inventory
- [ ] Queuing
- [ ] Network flow

Creative Ahmad Pk

Question # 10 of 10 (Start time: 09:48:31 PM, 08 March 2022) Total Marks:

To convert an Assignment problem into a maximization problem, which of the following operations would have to apply?

Select the correct option

- [ ] Deduct smallest element in any row form all other elements of the row.
- [ ] Deduct all elements of the row from highest element in that row
- [x] Multiply all the elements of assignment matrix by '-1'
- [ ] Deduct smallest element in the matrix from all other elements.

Creative Ahmad Pk

In a queue, the arrival pattern can be expressed in terms of --------.

- exact continuous variables
- number of servers
- [redacted] Probabilities
- customer behavior

Creative Ahmad Pk

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

Select the correct option

- False
- [redacted] True

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Creative Ahmad Pk

If a Transportation problem has 'm+n' constraints then how many number of unknowns it will have?

Select the correct option

- m/n
- m.n
- m+n
- m-n

Creative Ahmad Pk

In the assignment problem, the cost matrix is ----------.

Select the correct option

- rectangular
- square
- singular
- non-singular

Creative Ahmad Pk

The assignment problem is unbalanced if the cost matrix is not a ----------matrix.

Select the correct option

- [x] square
- [ ] singular
- [ ] rectangular
- [ ] non-singular

Creative Ahmad Pk

Which of the following represents "Arrival-->Service-->Service-->Service-->Out-->"?

Select the correct option

- [ ] Single channel multi phase system
- [ ] Multi channel multi phase system
- [ ] Single Channel Single Phase system
- [x] Multi channel single-phase system

N.C

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In the assignment problem, the cost matrix is ----------.

Select the correct option

- ○ rectangular
- ● square
- ○ singular
- ○ non-singular

Creative Ahmad Pk

Which of the following is the correct assumption for replacement policy when money value does not change with time?

Select the correct option

- ○ No Capital cost
- ○ No scrap value
- ● Constant scrap value
- ○ Zero maintenance cost



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While solving an Assignment problem by Hungarian's method, in the modified cost matrix if the minimum number of horizontal and vertical lines to cover zeros are not equal to the number of rows (or columns), then which of the following operation is done?

Select the correct option

- [x] Subtract smallest element of uncovered rows from all other elements of uncovered cells
- [ ] Subtract the smallest element from the next highest number in the element.
- [ ] Add smallest element of the uncovered cells to the elements to the line
- [ ] Subtract the smallest element from the element at the intersection of horizontal and vertical line.

Creative Ahmad Pk

There always exists a Basic Feasible Solution of a Balanced Transportation Problem.

Select the correct option

- [x] True
- [ ] False

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Which of the following property dictates that an Assignment matrix is a square matrix?

Select the correct option

- ○ Because we find row and column cost matrices
- ■ Because the allocations in assignment problem are one to one
- ○ Because we have to add the minimum uncovered element to the element at the intersection of horizontal and veridical lines
- ○ Because one to make allocations, one has to draw horizontal and veridical lines.

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Question # 10 of 10 (Start time: 12:15:22 AM, 09 March 2022) Total Marks

Suppose that a taxi firm has four taxis (the agents) available, and four customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the

Select the correct option

- ○ transportation
- ○ balanced assignment
- ■ unbalanced assignment
- linear programming

Creative Ahmad Pk

When the total allocations in a transportation model of m*n size is not equals to "m + n – 1", the situation is known as:

Select the correct option

- None of the above
- Degeneracy
- Unbalanced situation
- Tie situation

Creative Ahmad Pk

Suppose that a taxi firm has four taxis (the agents) available, and three customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the cost of picking up a particular customer will depend on the time taken for the

Select the correct option

- linear programming
- balanced assignment
- transportation
- unbalanced assignment ✓ Correct

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